Detection of Cosmic Ray Muons with a Coincidence Telescope

Minh Nguyen, Cirdan Klindworth, Austin Efflandt, Ian De Wall, and Mark Prince

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Abstract

Cosmic Ray Muons originate from cosmic rays colliding with atoms in Earth's upper atmosphere, creating a shower of particles. Muons are an unstable negatively-charged lepton, two hundred times more massive than an electron. In order to detect these high-speed particles, a scintillator is used as a target and a photomultiplier rests underneath. The scintillator creates photons when one of its atoms is struck by radiation; the photomultiplier makes use of multiple diodes in succession to cascade the small number of photons into a detectable amount of light. This system is applicable to multiple particle detection experiments because it outputs a pulse of energy related to the incident particle. In our case, we utilize a coincidence system with two detectors, in that both detectors must get a spike within a strict time window to be considered one signal. This allows us to ensure the incident particle is a muon and introduces flexibility in the angle we accept particles from. The greater the separation of our two detectors, the more narrow the acceptance angle. We intend to assess our measurements of muon count rates at various angles and evaluate any agreeability with existing models for cosmic ray muon count rates.

1 Introduction

1.1 Cosmic Ray Muons

A muon is a fundamental subatomic particle classified as a lepton in the Standard Model. It has the same charge as an electron (-1.602×10^{-19} C) but is around 200 times more massive (105 MeV/c^2). A cosmic ray muon, or CRM, is a secondary particle resulting from the collision of cosmic rays with atoms in Earth's upper atmosphere. The process in which secondary particles are created in a shower from the intervention of a primary cosmic ray is shown in **Figure 1**. At rest, muons have a lifetime of 2.2 μs.[1] However, because they are moving at relativistic speeds, which are a considerable fraction of the speed of light, CRMs experience ample time dilation. This means they can survive their trajectory to Earth's surface and get detected. Typically, CRMs have kinetic energies around 4-6 GeV, which correspond to velocities within a thousandth of a percent the speed of light. We obtain the energy-related statistics about these particles using the following relativistic equations:

$$v = c\sqrt{1 - \frac{1}{\gamma^2}}$$

$$E_{total} = T + mc^2$$

where γ is the Lorentz factor. For muons with kinetic energy of 4 GeV, the Lorentz factor is roughly 38.1, while muons with kinetic energy of 6 GeV have a Lorentz factor of 58.1.

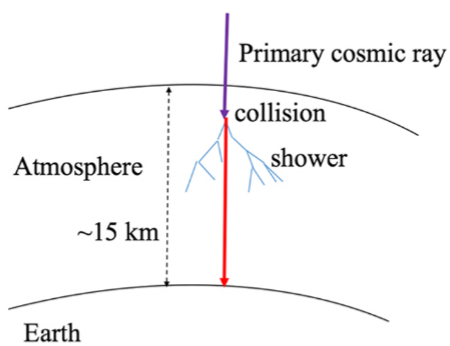

Figure 1: Illustration of cosmic rays facilitating a shower of secondary particles in the upper atmosphere.[1]

1.2 Theoretical Overview

Our objective is to be able to obtain a count rate for atmospheric muons that will reach our detector. The detection of cosmic ray muons involves the use of a photomultiplier and a scintillator. As the muon reaches our equipment, it will strike a scintillator. A scintillator creates light when a charged particle hits one of its atoms. Because this light is only a few photons in strength, we amplify the signal with an avalanche photodiode. Also known as a photomultiplier, its function is to strip off more photons from silicon within the photodiode. The photomultiplier uses a voltage to create an electron-hole pairing within the system, which allows the photons coming in to strip them off and create an "avalanche" of photons. This results in a stronger signal that is more easily read by our equipment. The signal from a detected muon will have an amplitude of 10-100 mV lasting roughly 0.5 μs, visually characterized by a steep rise and a long decaying tail.[1]

According to "Cosmic Ray Counting"[2] by D. B. Pengra, if we restrict the angle to straight down from the zenith, then the accepted intensity of cosmic ray flux is

$$I_v = 0.82.10^{-2} cm^{-2} s^{-1} str^{-1} \tag{1}$$

If we vary the angle θ with respect to the zenith, then a theoretical relationship between the intensity and the angle is:

$$I(\theta) = I_v \cos^2(\theta) \tag{2}$$

We use a 5cm x 5cm scintillator, yielding a total area of 25 cm^2. This means that, for a single detector, the expected intensity is $12.3\cos^2(\theta)$ counts per minute for a full 180° above the horizon.

2 Instrumentation

2.1 Scintillators and Silicon Photomultipliers

For this project, we implement the Saint Gobain BC 408 Scintillator and the SiPM 60035 SMA board for our photomultiplier, with a built-in high pass filter. **Figure 2** below shows the physical setup without the circuitry, and the step-by-step process of the device.

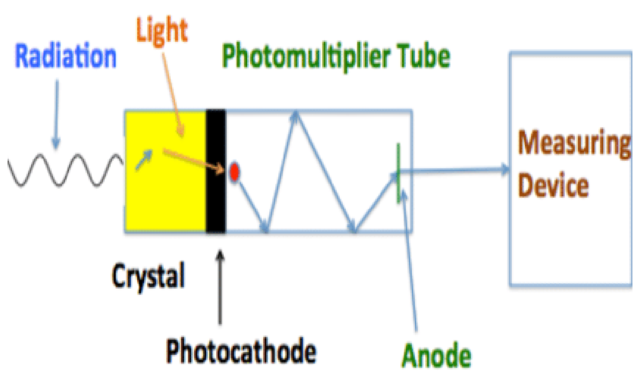

Figure 2: Step-by-step diagram for scintillator-photomultiplier design.[3]

2.2 Computerization and Circuitry

We would like to use an Arduino Uno in order to acquire and digitize the data. There are two issues. First, the raw signal from a muon is too short for the Arduino to detect, which has a minimum delay time of 1 millisecond. Second, the raw signal is too small for the Arduino digital reader to identify. For transistor-transistor logic, the input signal is defined as 'low' when between 0 V and 0.8 V with respect to the ground terminal, and ‘high’ when between 2.2 V and 5 V.[4] Because we expect a cosmic ray muon to produce a pulse in the range of 10-100 mV, the Arduino will read a muon detection as “low” even with the highest spikes in voltage. Thus, we would need to implement an amplification circuit and a signal stretcher in order for the Arduino to detect the muons.

The first step is to design a ‘high-pass’ filter connected to the SiPM. We use a circuit board that houses three 49.9 Ω resistors and four 10 nF capacitors, which are all included in the SMA-Biasing Board. A schematic of the board is shown in **Figure 3**.

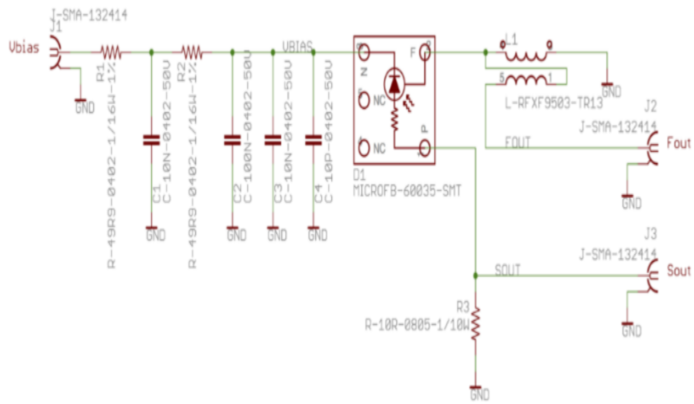

Figure 3: Board schematics for the MicroFC-SMA-60035-SMT.[5]

We use a reverse bias voltage of 29.4 V, as indicated in the reference design sheet for the SiPM. As we proceed to test this photomultiplier, it is important to recognize its sensitivity to light when this voltage is supplied. All testing will be done in a dark space.

3 Experimentation

3.1 Initial Tests

To test the functionality of our SiPM, we simply observe how it reacts to a flashlight without any scintillation involved. With the success of this experiment, we ensure the photomultiplier will properly receive light from a scintillator in the next step. As expected, we see the likeness of a $\frac{1}{x^2}$ relationship between voltage and distance of the light source, indicating a spherical emission from the source (see **Figure 4**). In this section of the experiment, we made use the Arduino Uno’s analog-to-digital converter, or ADC. The unit employed by the software is in “ADU” which is an arbitrary analog-to-digital value directly related to voltage. Nonetheless, the relationship reflects our expectation and confirms the functionality of the SiPM.

Moving onto muon detection, we place one scintillator wrapped in reflective foil over the SiPM in a controlled dark space. A hole in the wrapping is cut out to be just large enough for the area of the SiPM. The reflective foil decreases the chance that photons will escape the scintillator before entering the SiPM. Darkness is necessary due to the photodiode’s high sensitivity; any excess light may create unwanted noise.

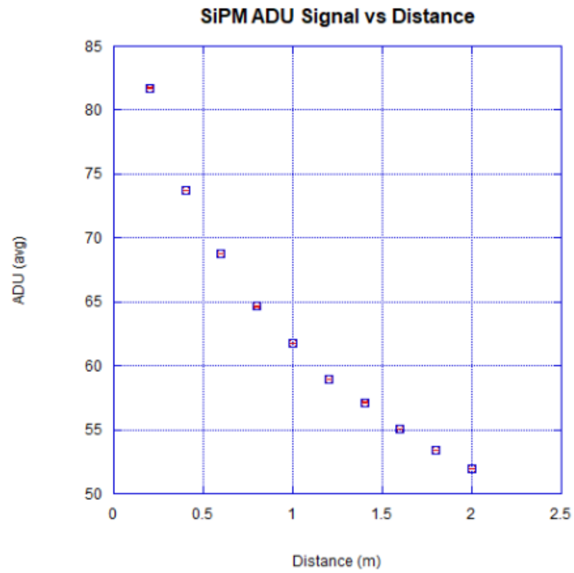

Figure 4: We test the functionality of our SiPM with a flashlight at varying distances.

Assuming time is represented in the horizontal axis and voltage on the vertical, an oscilloscope should display spikes around .5 μs wide and 10-100 mV high to indicate a muon disturbing the scintillator.[1] This type of pulse is displayed on our oscilloscope in **Figure 5**.

We wish to experiment with a single detector first; this involves varying the angle our scintillator is positioned relative to zenith. We initially suspected direct proportionality between the count rate and the effective cross sectional area of exposed scintillator. While we do not intend to extensively experiment with a single detector, we are curious about a potential relationship between this angle and count rate. Because muons plummet in from all angles, the orientation of a single detector does not actually correlate to a meaningful difference in exposed cross sectional area due to the three-dimensional nature of the scintillator. Therefore, no convincing trend relating angle and count rate is concluded with a single detector, especially not the $\cos^2(\theta)$ model (see **Figure 8**).

3.2 The Coincidence Method

For a single detector, particles from all angles are captured and detected by the photomultiplier. Therefore, in addition to muons, we see background noise as well as other particles. With two detectors in coincidence, we can distinguish muons coming from a certain range of angles and filter out most other particles. A muon count is only confirmed if both detectors manage to detect it simultaneously. We decide to have our two detectors placed 80 cm apart, which means our acceptance angle is $2\alpha = 10.1°$: the angle where one muon can strike both scintillators along its trajectory. Any other muons or particles outside of this angle are rejected by the coincidence system. As we run experiments with this setup, we vary the angle of the telescope from 0° to 90° in 10° increments. We understand that each trial will accept muons at $\pm$ 5.05° due to the acceptance angle. For example, the data taken at $\theta = 60°$ relative to zenith will include muons with incidence angles ranging from 54.95° to 65.05°.

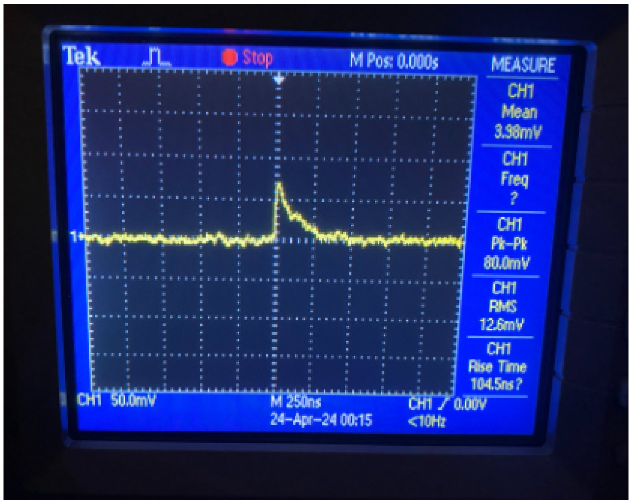

Figure 5: A voltage spike due to a CRM is observed on an oscilloscope screen.

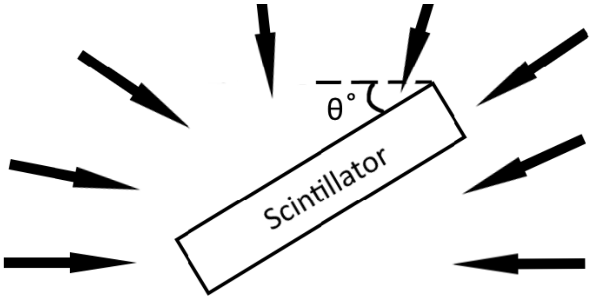

Figure 6: A simple representation of incident muons impacting a single scintillator. Even with an angled orientation, muons impact the surface from horizon to horizon.

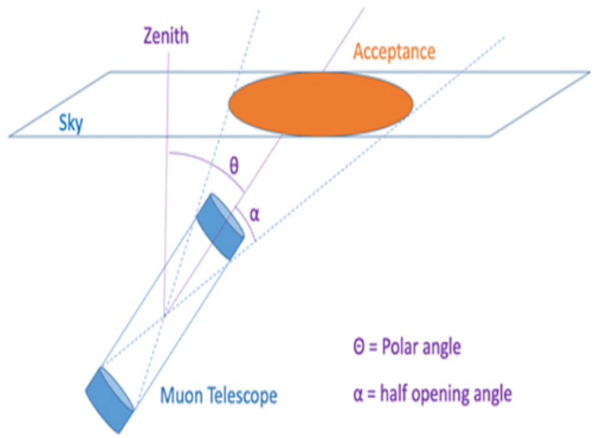

Figure 7: An illustration of a coincidence telescope's geometry.[1]

3.3 Considering Sources of Noise

Aside from muons, there might also be other components picked up by the detector, leading to a false muon detection.

Electronic noise: One source of noise is from our electronic setup, where instabilities in equipment might introduce noise into the signal, leading the oscilloscope to pick up additional peaks that are not related to muon interactions. These noises are often characterized by chaotic fluctuations in the oscilloscope, as opposed to a single peak in the case of a radioactive source.

Saturation: If the signal amplitude exceeds the dynamic range of the oscilloscope due to an extended duration of exposure of the photomultiplier to some sources of light, the read out from the oscilloscope might be larger than the expected signal. This might explain the occasional high peaks in our oscilloscope which exceed the upper range for a muon's energy, sometimes surpassing 200mV.

Background radiation: There might be other sources of radiation present in the surrounding areas. We experimented with a Strontium radioactive source and a Uranium glaze, and despite ensuring that these sources are not in close proximity with our equipment, there is no guarantee that the detectors won't pick them up due to its high sensitivity. In addition, there may also be other sources of high energy cosmic particles such as protons or atomic nuclei, as well as radiation coming from humans, lab instruments, and building materials. There are also additional radiation coming from underground, which might affect our count rate if we are not shielding our detectors from beneath.

4 Analysis of Results

4.1 Single detector

To test how well our detector agrees with the theoretical model, we use one detector and vary it from 0 to 90°, with an increment of 15°. We measure the counts that we get in two minutes. The result is shown in **Figure 8**.

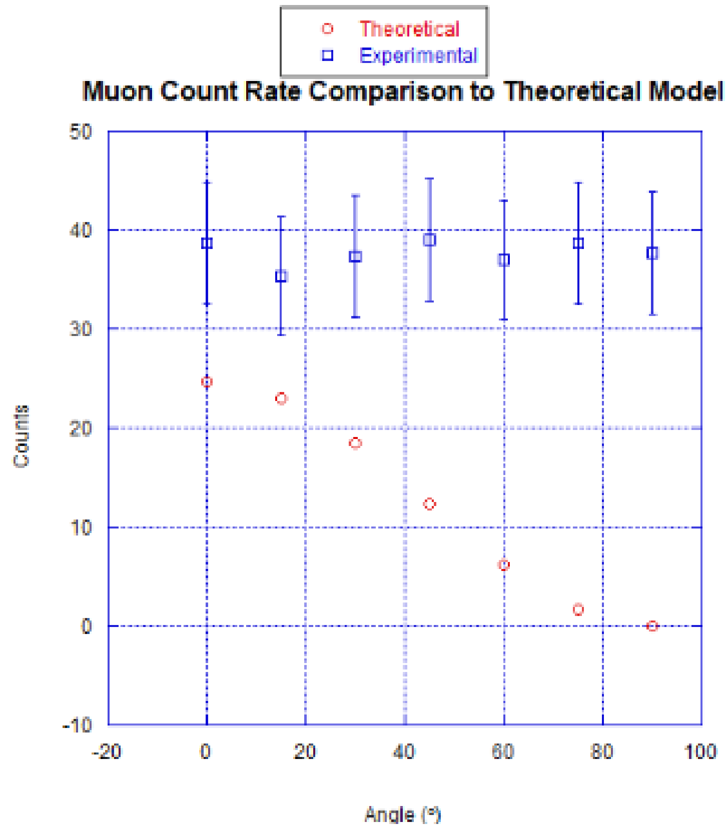

Figure 8: Single detector measured vs theoretical count rate/two mins.

It can be seen that our measured count rate does not vary with angle, to test how well our results fit the $cos^2(\theta)$ model, we calculate the chi-squared value of our 7 data points:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 48.9 \tag{3}$$

With 7 data points, and two free parameters, the fit test and the number of counts, we would have 5 degrees of freedom. So our value for chi-squared is $\chi^2 = \frac{48.9}{5} = 9.78$ per degree of freedom. This means that we have less than 0.05% chance that our measured result agrees with the theoretical model. Therefore, for a single detector, there is little evidence to suggest that the count rate follows our theoretical model of angle dependence.

4.2 Coincidence Detectors

When putting two detectors in coincidence with 80 cm separation, we would expect to see a count rate of 1.403 counts/min for an acceptance angle of 10.1° if we were to point the detectors directly at the zenith. Thus the theoretical count rate is:

$$1.403cos^2(\theta) \tag{4}$$

We take an increment of 10° from 0 to 90, with an addition of 35°, and plot the theoretical counts vs the measured counts, as shown in **Figure 9**. Note that 0° means the detectors are pointing at the zenith, and 90° is the detectors being parallel to the horizontal.

The measured count rate is decreasing with angle, suggesting that there is some angle dependence for the coincidence detectors. The measured counts is lower than expected for angles less than 50 degrees, and higher than expected for angles above 60 degrees. The model suggests that we don't get as many muons for lower angles and more for higher angles. To see how consistent our measured values are with the theoretical model, we calculate the chi-squared value for all 11 data points, which yields 2.18.

We have two free parameters, and 11 data points, so that yields 11-2 = 9 degrees of freedom. Thus our chi-squared per degree of freedom is $\frac{2.18}{9} = 0.242$. This would yield almost 100% chance that our measured results agree with the theoretical model. However, since we only take data in a short period of 5 minutes and with few trial runs, our data is insufficient to make a robust conclusion about the consistency with the model. In addition, we are using our eyes to detect the muons, which would be less reliable than using software, so there is a factor of human error. Any spikes we view that are only displayed on one oscilloscope we assume are muons coming from outside the acceptance angle or noise. We also would like to find a way to refine the timing, rather than guessing what "simultaneous" spikes should look like with an oscilloscope's unreliable refresh rate.

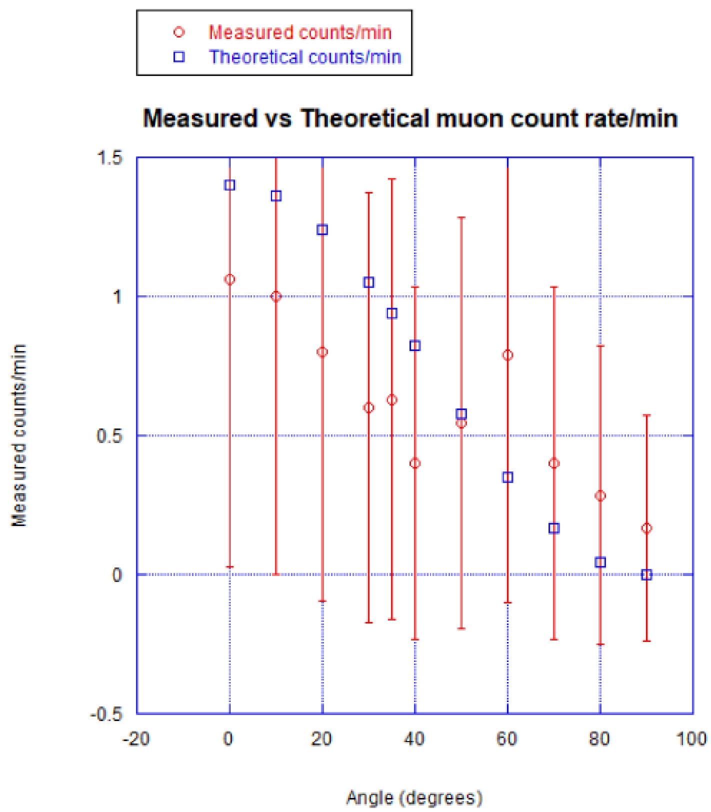

Figure 9: Coincidence detectors measured vs theoretical count rate/min.

4.3 Discussion

Given the opportunity for further research, we would like to digitize our muon counting process by using an Arduino Uno to count the number of simultaneous events detected by both detectors in our coincidence setup to have a more precise and scientifically relevant experimental result. The process we considered using for the future was to attach an exterior circuit to our detectors that would allow for a longer decay or a stretching of the detector signals. In our experiment's current configuration, the signal that our detectors are producing has very sharp and quick decays which end up being to short for the Arduino Uno system to be able to process. The addition of the signal stretching circuit to our experiment allows for the detector signals to decay more slowly and with a more pronounced tail end, enabling the Arduino Uno system to process the signals and in turn count the simultaneous events detected by both detectors. For future experimentation, the use of different connectors or a different PCB board for our detectors could potentially allow for the addition of an external circuit that could stretch the detector signals and allow for the digitization of the counting of simultaneous detected events, but the exact connectors or PCB boards to be used require further research and consideration.

References

[1] Smith, Walter F. “Chapter 21: Nuclear and Particle Physics.” *Experimental Physics: Principles and Practice for the Laboratory.*

[2] Pengra, D. B, “Cosmic Ray Counting”, Nov. 2021,

[3] Meng, Fang. ”Development and Improvement of Cerium Activated Gadolinium Gallium Aluminum Garnets Scintillators for Radiation Detectors by Codoping.” 2015.

[4] Fairchild SemiConductor TM. ”DM7490A Decade and Binary Counter”. 1986.

[5] Onsemi. ”Reference Designs for the SMA and SMTPA MLP-Packaged SiPM Evaluation Boards”.